



Day 4: Data Modeling Fundamentals

Masterclass Series: Learn how to structure and optimize enterprise-grade data models in Power BI professionally.

The Isolated Tables Problem

❗ Disconnected Silos

Imagine a retail business maintaining three distinct tables:

- 👤 **Customers Table** (Saves demographics)
- 🛒 **Orders Table** (Tracks purchase lines)
- 📦 **Products Table** (Contains inventory detail)

📊 Can Power BI Analyze?

If all these data silos remain completely disconnected, can the analysis work?

No. Unconnected tables cannot interact with each other.

"Tables must communicate. That communication is called a Relationship."

What is Data Modeling?

Core Definition

Data Modeling is the structured blueprinting process of organizing, mapping, and connecting multiple database tables so they function as a unified analytical engine.

Organizing connections systematically prevents query failures, reduces data size, and speeds up report generation.

The Family Analogy

Think of your database tables like members of an extended family.

"If family members are unrelated, communication becomes difficult. Relationships connect them together, allowing seamless communication."

Active Datasets Today

 Customers

CustID	Name	City
C101	Rahul	Hyderabad
C102	Sneha	Chennai
C103	Arjun	Bangalore

 Orders Fact

OrderID	CustID	ProdID	Amt
O001	C101	P201	5000
O002	C102	P202	7000
O003	C101	P203	4000

 Products

ProdID	ProdName	Cat
P201	Laptop	Elect.
P202	Printer	Elect.
P203	Mouse	Access.

Interactive Class Scenario 1: E-Commerce

Dim_Customers (Master List)

Primary Keys (Unique)

Rahul	C101
Sneha	C102
Arjun	C103

Fact_Orders (Transactions)

Foreign Keys (Duplicates)

Order O001 - Laptop	Waiting...
Order O002 - Mouse	Waiting...
Order O003 - Printer	Waiting...

Assign Rahul's orders (C101)

Assign Sneha's orders (C102)

⚙️ SELECT A MASTER CUSTOMER ABOVE OR ASSIGN THEIR ORDERS
Reset Grid

Interactive Class Scenario 2: Healthcare System

Dim_Patients (Master List)

Primary Key (PK)

Amit Kumar (Blood Group: A+)	P801
Zara Khan (Blood Group: B-)	P802

Fact_Visits (Clinic Visits Logs)

Foreign Key (FK)

Visit V55 - Dr. Roy (Cardiology)	Unlinked
Visit V56 - Dr. Mehta (General)	Unlinked
Visit V57 - Dr. Roy (Cardiology)	Unlinked

Link Patient Amit (P801) to all visits

Link Patient Zara (P802) to all visits

CLICK A BUTTON TO MAP PATIENT IDS DYNAMICALLY

Clear links

Star Schema vs. Flat Table

The Star Schema Standard

A star schema organizes data by placing a central **Fact Table** (which stores numeric transactions) at the core, connected outward to surrounding **Dimension Tables** (which store descriptive lookup data like customers or products).

Key Advantages: High-speed query performance, highly organized structure, and clean filter propagation.

The Flat Table Alternative

A single massive table containing all transaction records and descriptive text. Causes large file sizes, slow queries, and data duplicates.



Power BI Steps 1-4: Create First Relationship

1 Load Your Sources

Go to Home -> Get Data -> Excel. Select your Excel tables: Customers, Orders, Products.

2 Navigate to Model View

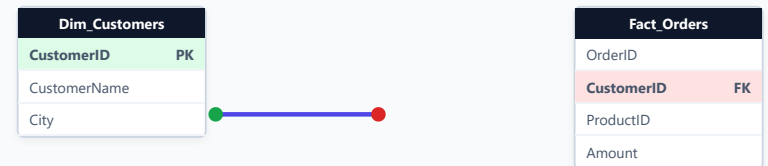
Click the
Model View
tab (the third network diagram icon on the left vertical pane).

3 Locate Key Fields

Identify CustomerID (PK) in Customers and CustomerID (FK) in Orders.

4 Drag and Link Keys

Drag CustomerID from Customers and drop it directly on CustomerID in Orders.



Active link created. Hovering over this line highlights the key fields.

Power BI Steps 5–8: Complete Star Schema

5 Link Products Table

Locate the Products table. Identify ProductID (PK) in Products and ProductID (FK) in Orders.

6 Drag and Link Products

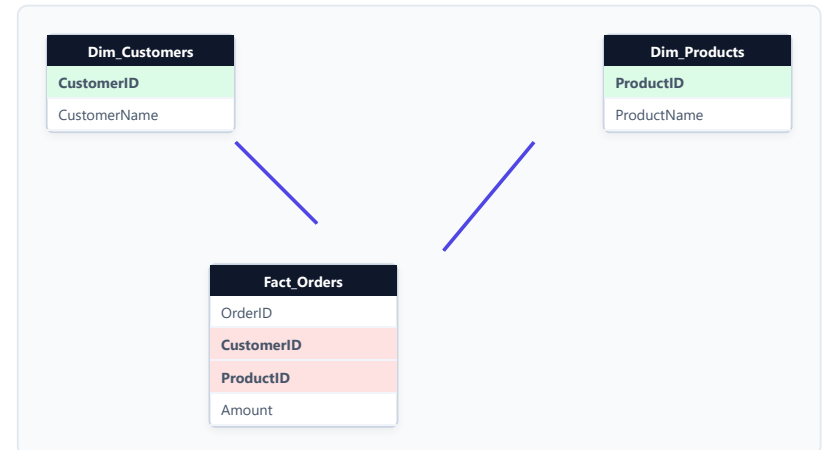
Drag ProductID from Products and drop it directly on ProductID in Orders.

7 Observe the symbols

Verify that both lines display the
1 to Many (1 → *)
cardinalities and pointing filter arrows.

8 Verify Your Design

Add CustomerName, ProductName, and Amount to a single table visual in Report view to confirm they display correctly.



★ Star Schema completed. The Fact table is at the core, surrounded by Dimension tables.

Modeling Best Practices

✓ Model To-Dos

- ✓ Always use a Star Schema structure.
- ✓ Keep table schemas streamlined.
- ✓ Use clear, business-friendly column names.
- ✓ Verify active filter pathways.

✗ Modeling Pitfalls

- ✗ Avoid loading large flat files directly.
- ✗ Avoid building redundant lookup tables.
- ✗ Do not leave tables disconnected.
- ✗ Watch out for slow visual loading times.

"A solid data model ensures fast, responsive, and accurate dashboards."

Hands-on Practice

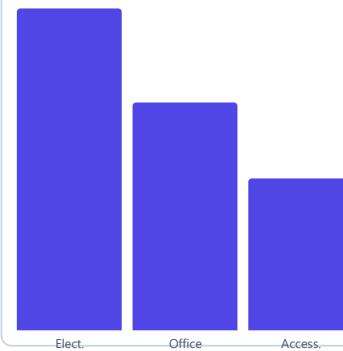
Student Exercises

- ✓ Task 1: Link Customers and Orders tables in your file.
- ✓ Task 2: Link Products and Orders tables in your file.
- ✓ Task 3: Build a Column chart showing sales by product category.
- ✓ Task 4: Create a table visual to find the highest-value customer.

Practice Model Output Visuals

[LINKED OK](#)

Sales Revenue by Category



Top Purchasing Customers

Sneha	\$7,000
Rahul	\$5,000

Common Doubts & Session Wrap-up

❓ Standard Modeling Doubts

Q: Why are model relationships needed?

👉 To connect tables for unified data analysis.

Q: What happens if tables remain disconnected?

👉 Your visuals will return blank cells or calculation errors.

★ The Data Standard

"Without relationships, data is just isolated files.
With relationships, data becomes meaningful
analysis."

Key Takeaway: Clean data is important... but connected data
creates intelligence.